

Project Design Phase-II

Data Flow Diagram & User Stories

Date	01 October 2026
Team ID	6ab6292ddcdae33acb3906da
Project Name	Import Data Using Transform Maps
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. For this project it shows how employee training data moves from a spreadsheet into ServiceNow through an Import Set and a Transform Map, how the employee department is pulled in by dot-walking, and how roles and Access Control Lists (ACLs) control who can see the data.

Level 0 (context diagram)

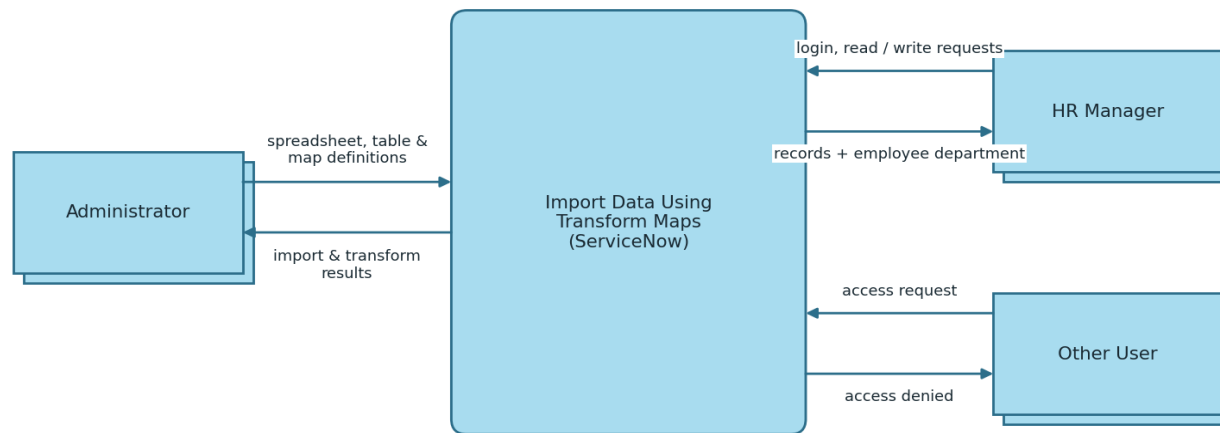


Figure 1: DFD Level 0

Level 1: Importing data using a transform map

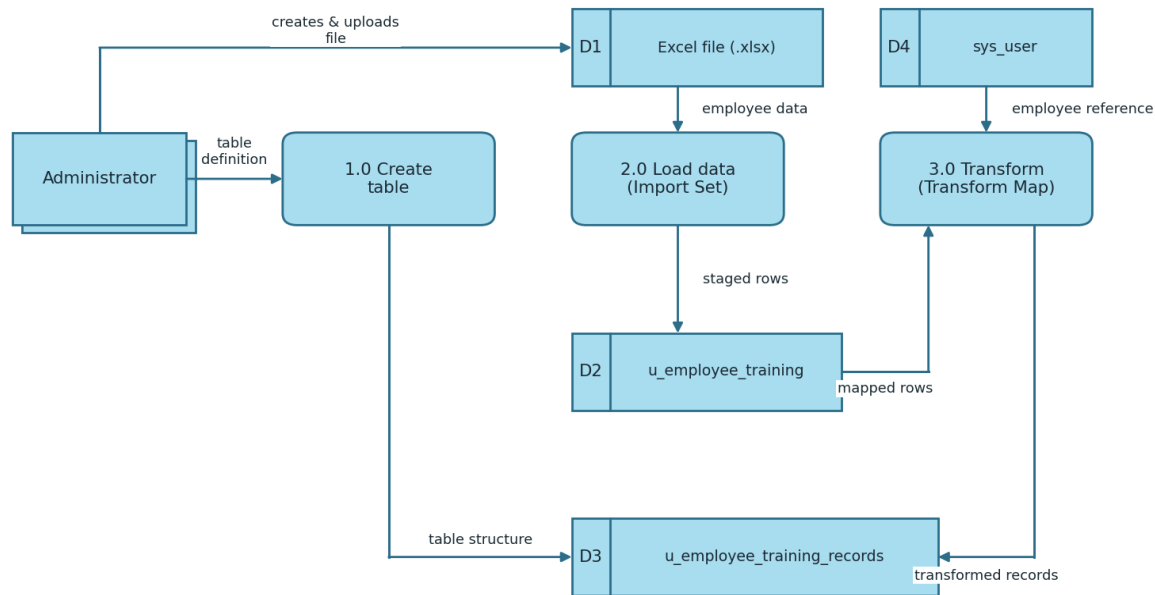


Figure 2: DFD Level 1 (import pipeline)

Flow

1. Administrator creates the Employee Training Records table (1.0) with Training Name, Completion Date, Status and Employee fields.
2. Administrator prepares the .xlsx file (D1) and uploads it using Load Data (2.0); the rows are staged in the import set table u_employee_training (D2).
3. Administrator creates the transform map and field maps; the sys_user table (D4) is used to resolve the Employee reference.
4. Running the transform (3.0) moves the mapped rows into u_employee_training_records (D3).

Level 1: Dot-walking and securing data with ACLs

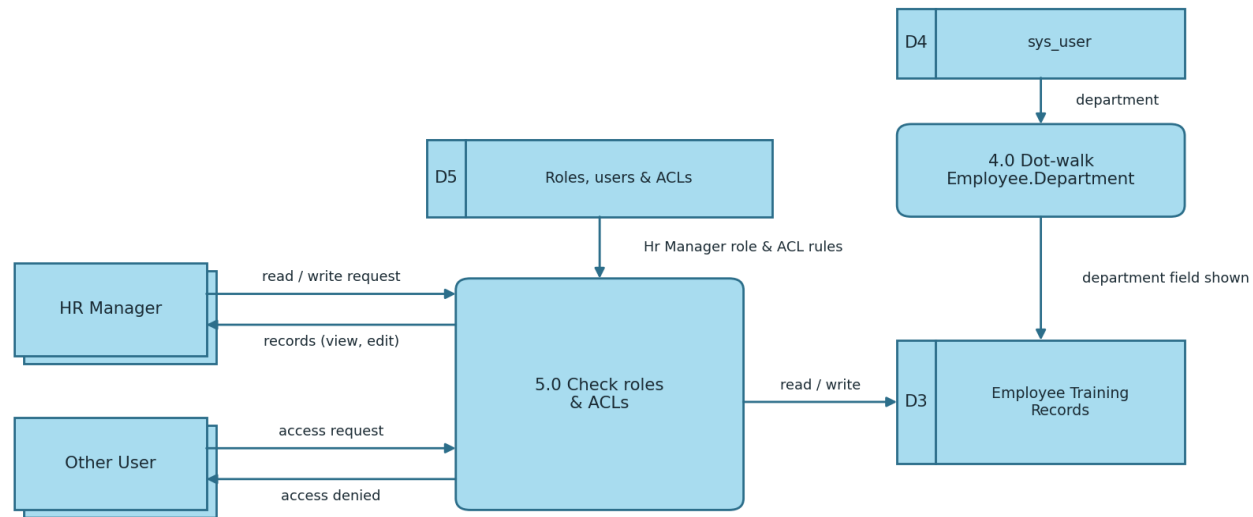


Figure 3: DFD Level 1 (dot-walking and access control)

Flow

- Dot-walking (4.0) reads Department from the linked sys_user record and shows it on the Employee Training Records form and list.
- When a user requests the table, the roles and ACL rules (D5) are checked (5.0).
- A user with the Hr Manager role is granted read and write access; any other user is denied access.

User Stories

User stories for the Import Data Using Transform Maps project, grouped by user type and functional requirement (epic).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Table creation	USN-1	As an administrator, I can create the Employee Training Records table with the fields Training Name, Completion Date, Status and Employee (reference to sys_user).	Table u_employee_training_records is created with all four columns	High	Sprint-1
		USN-2	As an administrator, I can add choice values (In Progress, Completed) to the Status field.	Status drop-down shows the configured choices	Medium	Sprint-1
	Data import	USN-3	As an administrator, I can prepare an .xlsx file with Training Name, Completion Date, Status and Employee columns.	File has a header row and the records to import	High	Sprint-1
		USN-4	As an administrator, I can upload the file using Load Data to create the import set table u_employee_training.	Import progress shows State: Complete and Completion code: Success	High	Sprint-1
	Transform map	USN-5	As an administrator, I can create a transform map from the import set table to the Employee Training Records table.	Transform map is saved with the correct source and target tables	High	Sprint-1
		USN-6	As an administrator, I can add field maps for training name, completion date, status and employee.	Field Maps list shows all four fields mapped	High	Sprint-2
		USN-7	As an administrator, I can run the transform to move staged rows into the target table.	Transformation shows Complete / Success and records appear in the target table	High	Sprint-2
	Dot-walking	USN-8	As an administrator, I can dot-walk Employee.Department onto the form layout.	Department field is visible on the form and in the list view	Medium	Sprint-2
	Roles & users	USN-9	As an administrator, I can create a user (backup user) and an Hr Manager role, and assign the role to the user.	Roles tab of the user shows Hr Manager as Active	High	Sprint-2
		USN-10	As an administrator, I can restrict the application menu and module to the Hr Manager role.	Menu and module Roles field lists Hr Manager	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
	ACL	USN-11	As an administrator, I can elevate to security_admin and create a read ACL on the table for the Hr Manager role.	ACL (operation: read) exists with Hr Manager under Requires role	High	Sprint-3
		USN-12	As an administrator, I can create a write ACL on the table for the Hr Manager role.	ACL (operation: write) exists with Hr Manager under Requires role	High	Sprint-3
HR Manager	Record access	USN-13	As an HR manager, I can open Employee Training Records and view and edit the records.	When impersonated, the list opens and the fields can be edited	High	Sprint-3
		USN-14	As an HR manager, I can see each employee's department in the record without entering it.	Department is shown from the linked sys_user record	Medium	Sprint-3
Other User	Record access	USN-15	As a user without the Hr Manager role, I cannot see the Employee Training Records table.	When impersonated, the application is not in the navigator	High	Sprint-3